

Varshini Akula

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SUMMARY

A software engineer with proven technical experience from exposure at a leading global company, adept at solving complex business challenges through innovative software engineering and AI-driven solutions.

EDUCATION

Arizona State University

Master of Science in Computer Science

Tempe, AZ

Aug 2025 - May 2027

Jawaharlal Nehru Technological University

Bachelor of Technology in Computer Science and Engineering (Data Science) (9.49/10 CGPA)

Hyderabad, India

May 2024

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, HTML, CSS, SQL, C, Bash

Tools/Frameworks: FastAPI, Maven, Git, GitHub, Bitbucket, JIRA, Jenkins, Postman, Swagger, Docker, Kubernetes, Spring, Spring Boot, REST, Apache Kafka, WebSocket, JUnit, Mockito, Next.js, Tailwind CSS, Linux, NVIDIA A100 GPU clusters

Machine Learning: PyTorch, TensorFlow, SciPy, pandas, scikit-learn, Hugging Face Transformers, LangChain, OpenAI APIs, Rasa NLU, RAG Systems, LLM Fine-Tuning

Databases: MySQL, MongoDB, PostgreSQL

Core Competencies: Software Requirements and Specifications, Cloud Computing, Object Oriented Programming, Data Structures and Algorithms, Database Management Systems, Natural Language Processing, Agentic AI

Certifications: AWS Certified Cloud Practitioner (Dec 2024)

PROFESSIONAL EXPERIENCE

JPMorgan Chase, Hyderabad, India: Software Development Engineer I

Jul 2024 – Aug 2025

- Designed and shipped a production full-stack conversational AI system using **RASA NLU, Python services, AWS, and React** for document storage onboarding, automating metadata extraction and **eliminating 320+ hours of manual data entry per quarter**.
- Built **Java, SQL, and AWS microservices** to automate document storage record CRUD operations across Consumer & Community Banking, **cutting storage provisioning time by 40%**. Engineered backend Java applications leveraging **MVC architecture and Spring Boot** to seamlessly integrate **microservices and REST APIs**.
- Hardened production observability with **structured logging, distributed tracing, and health-check endpoints** feeding **real-time monitoring dashboards** for services under a **99% uptime SLA**.

JPMorgan Chase, Hyderabad, India: Software Engineering Program Intern

Jan 2024 – Jun 2024

- Engineered a full-stack LLM-powered incident chatbot with **Python, LLMs, and TypeScript** that summarized tickets, generated severity-based runbooks, and **reduced mean incident query resolution time by 60%**.
- Developed a real-time data streaming application with **Java, Spring Boot and Apache Kafka** for the Global Technology & Innovation team, lowering cross-system propagation latency for **70+ downstream applications**.

PROJECTS

Hybrid Multi-Retriever Orchestration for Long-Term Conversational Memory, Research Project

Fall 2025

- Built a **RAG orchestration** system in Python, routing queries across **BM25, TF-IDF, FAISS, SVM, and time-weighted** retrievers using user intent-based and rule-based decision logic for long memory context queries.

- Improved **Recall@5** from **0.65 to 0.83 (+18 pts)** on **LongMemEval** with **Qwen2.5-7B** and **+4 pts on Locomo** with **LLaMA-3.1-8B** through **few-shot prompt engineering**.

Construction Project Intelligence

Spring 2026

- Built a full-stack **FastAPI & React** document intelligence platform with **RBAC, RAG Q&A** with citations, **PostgreSQL-backed indexing**, RFI classification, and real-time analytics dashboards; **reduced manual lookup time by 95%**.

FlowMind: Visual Design and Debug Tool for AI Agents

Spring 2026

- Integrated Building a visual AI workflow platform for designing, debugging, and deploying AI assistants with **graph-based conversation flows, RAG citations, live analytics dashboards**, and **hosted chat deployment** using **Next.js 15, React 19**, and **TypeScript**.

PUBLICATIONS

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- Evaluating the Performance of Fake Face Detection Methods in Forensics, IJITCE, Jun 2024
 - Optimizing Prediction of Security Data Using Feature Selection and Ensembling, IJITCE, May 2023

LEADERSHIP AND ACTIVITIES

ACM at ASU - Engagement Manager

Oct 2025 – Apr 2026

- Supported engagement and outreach strategy for ACM at ASU, a 4,700+ member computing student organization focused on networking, learning, mentoring, and technical community-building.

Literaria Clava, Literature Club of GNITS, Hyderabad - President

- Led 110+ members of GNITS' literary club, fostering student engagement in creative writing, public speaking, literary expression, and campus-wide communication initiatives.

Women's Leadership Conclave 2024, GNITS, Hyderabad - Student Chair

- Led student coordination for the STEM theme, supporting speaker engagement, session planning, and execution of a women's leadership forum focused on technology, innovation, and representation in STEM featuring **100+ speakers**.